



NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

Lecture 3: CA6121 AI in Project Management

Project Planning & Scheduling, Resource, Optimization, Prediction, Forecasting

Presented by Dr Ravish Maqsood





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OBJECTIVES

- Overview of planning & scheduling
- Schedule network diagram
- Dependencies
- Critical path method (CPM)
- Resources
- AI tools in Project Management
- Case studies:
 - AI in Project Management
 - AI usage in industry



Project Planning & Scheduling

Project Management



Planning Concept

Project Planning & Scheduling



Planning

A plan is used to:

- Align expectations of all parties involved
- Minimize workflow time of a project
- Minimize or avoid conflict
- Manage changes effectively
- Resource allocation
- Many more...



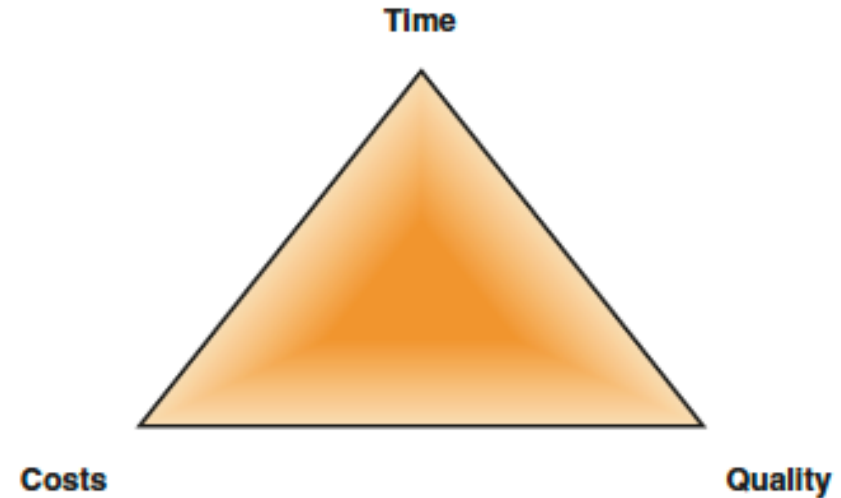
Planning helps to answer:

- **What?**
 - Identify and describe the subprojects, work packages and tasks; The result is the work breakdown structure.
- **Until when?**
 - Describe the timeline; The result is the time schedule and milestone plan.
- **Who/how many?**
 - Show which forces are to be used with which qualification, which personnel and material resources are needed. This is documented in the resource and cost plan. The project organization describes the type of cooperation.
- **Conditions?**
 - Specify the technical and spatial requirements, list the known restrictions. This is noted directly in the project plan.



Planning challenge to balance following three aspects

- **Quality**
- **Cost**
- **Time**





Project Planning and Scheduling

Project Planning & Scheduling



Key Processes

- 1. Define Activities**
- 2. Sequence Activities**
- 3. Estimate Activity Duration**
- 4. Develop Schedule**



Project Planning & Scheduling

Define
Activities

Inputs

.1 Project management plan

- Schedule Management Plan
- Scope baseline



Project Planning & Scheduling

Define
Activities

Inputs

Project Scope Baseline

- Project WBS,
- Deliverables,
- Constraints,
- Assumptions as mentioned in scope baseline document.



Project Planning & Scheduling

Define
Activities

Outputs

- a. **Activity List**
- b. **Activity Attributes** (location, cost, resource requirement etc.)
- c. **Milestone List**
- d. **Change Requests**
- e. **Project Management Plan Updates**
 - Schedule Baseline
 - Cost Baseline



Project Planning & Scheduling

Sequence
Activities

1. Define Activities

2. Sequence Activities

3. Estimate Activity Duration

4. Develop Schedule



Project Planning & Scheduling

Sequence
Activities

Tools & Techniques

- **Precedence Diagramming method**
 - Finish-to-start (FS)
 - Finish-to-finish (FF)
 - Start-to-start (SS)
 - Start-to-finish (SF)
- **Dependency Determination and Integration**
 - Mandatory dependencies
 - Discretionary dependencies
 - External dependencies
 - Internal dependencies
- **Leads and Lags**



Project Planning & Scheduling

Sequence
Activities

Outputs

a. Project Schedule Network Diagrams

b. Project Documents Updates

- Activity Lists
- Activity Attributes
- Milestone Lists
- Assumption log



Project Planning & Scheduling

Estimate
Activity
Durations

1. Define Activities

2. Sequence Activities

3. Estimate Activity Duration

4. Develop Schedule

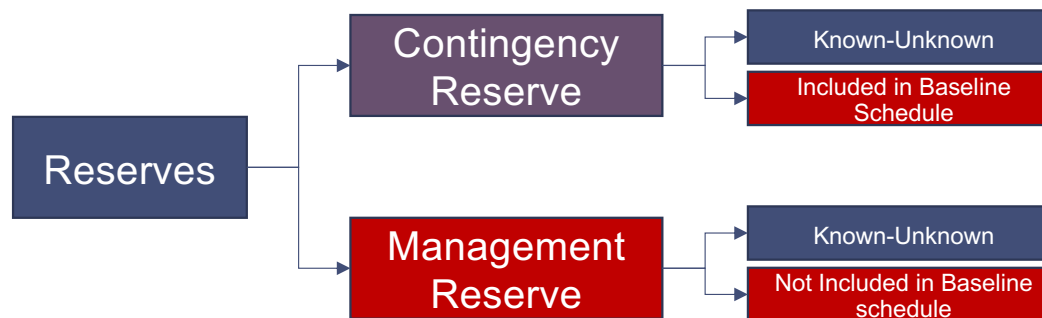


Project Planning & Scheduling

Estimate
Activity
Durations

Tools & Techniques

- **Reserve Analysis (Contingency)**
 - Known – unknowns
 - Buffers or reserves
 - Management Reserves are not included in schedule baseline





Project Planning & Scheduling

Estimate
Activity
Durations

Outputs

a. Duration Estimates

b. Basis of Estimates

- Documentation of the basis of the estimate development
- Assumptions
- Constraints
- Range of accuracy (+/- 10%)
- Confidence level of the final estimate
- Risks

c. Project Documents Updates

- Activity Attributes
- Assumption log
- Lessons register



Project Planning & Scheduling

Develop
Schedule

1. Define Activities
2. Sequence Activities
3. Estimate Activity Duration
4. Develop Schedule



Project Planning & Scheduling

Develop
Schedule

Tools &
Techniques

- **Refer to schedule development details in this lecture later.**



Project Planning & Scheduling

Develop
Schedule

Outputs

a. Schedule Baseline

- Activity duration estimates; quantitative assessment of required to complete an activity.
 - 2 weeks +/- 2 days.
 - 15% probability of exceeding given time.
- Agreed and approved version of Schedule.
- Any changes from now must be approved through Change Management Process.

b. Project Schedule

- Milestone Charts
- Project Schedule Network Diagrams

c. Schedule Data

- Resource requirements

d. Project Schedule Calendars

- Workdays, shifts available for the schedule



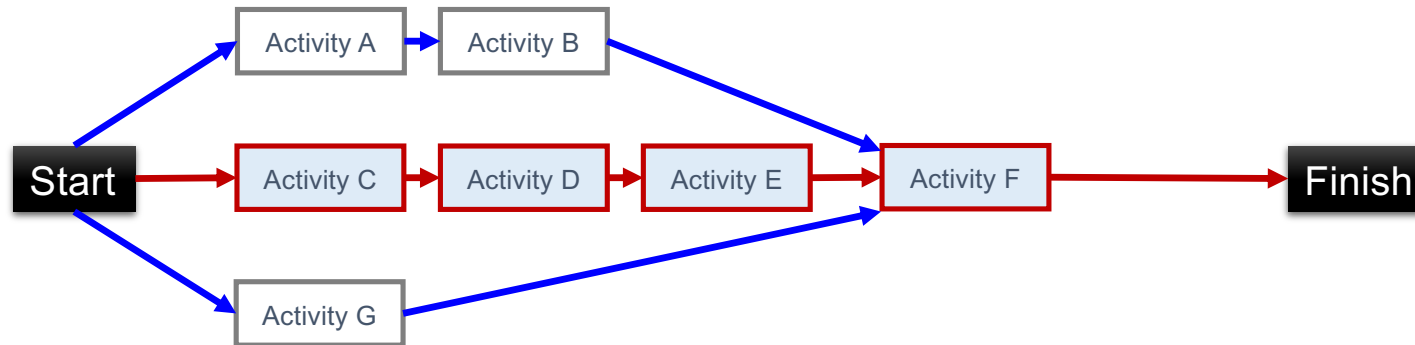
Schedule Network Diagram

Project Planning & Scheduling



Project Schedule Network Diagram

It's a graphical representation of the logical relationships and known as dependencies, among project schedule activities.



AI Tool can help



Dependencies

Project Planning & Scheduling



Dependency Determination and Integration

1. **Mandatory Dependencies:**

- Legally or contractually required.
- Hard Logic or Hard Dependencies.

2. **Discretionary Dependencies:**

- Preferred Logic, Preferential Logic or Soft Logic.
- Should be fully documented and reviewed in fast-track project environment as they may add to total float.

3. **External Dependencies:**

- Relationship between project activities and non-project activities.
- Outside project's team control.

4. **Internal Dependencies:**

- Inside project's team control.
- Activities are interdependent.

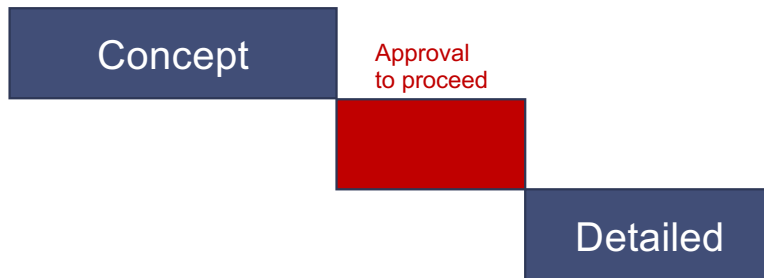


Dependency Determination and Integration

1. Mandatory Dependencies:

- Legally or contractually required.
- Hard Logic or Hard Dependencies.

Example:



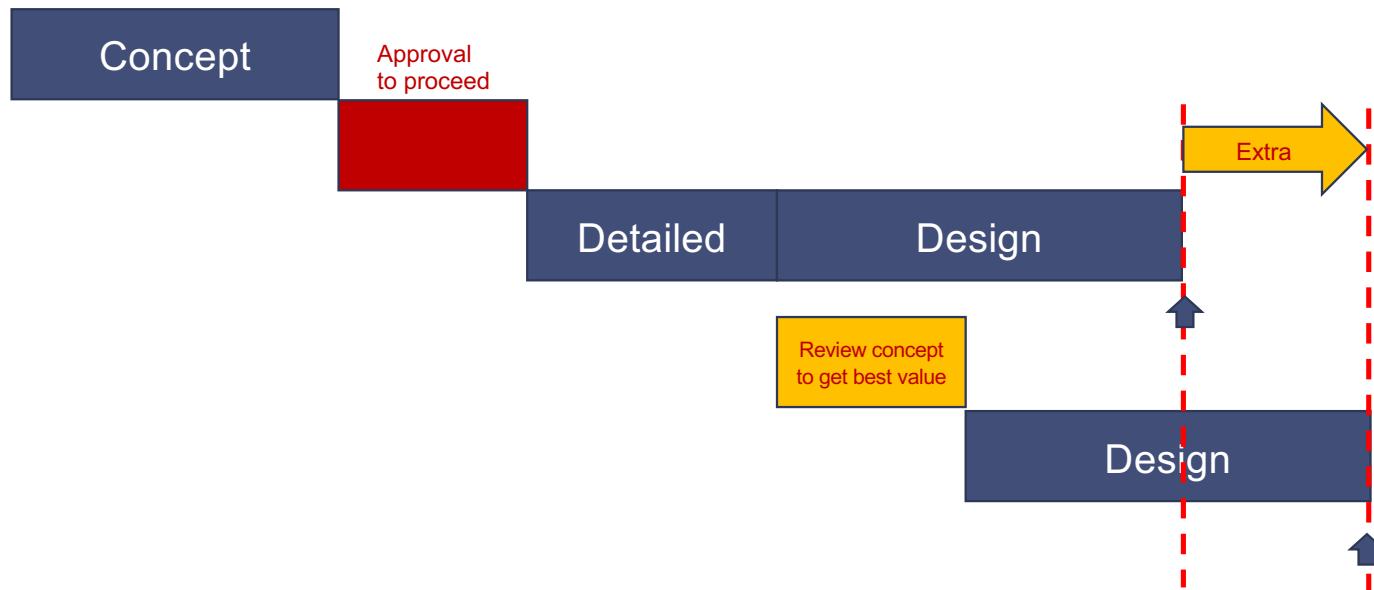


Dependency Determination and Integration

2. Discretionary Dependencies:

- Preferred Logic, Preferential Logic or Soft Logic.
- Should be fully documented and reviewed in fast-track project environment as they may add to total float.

Example:



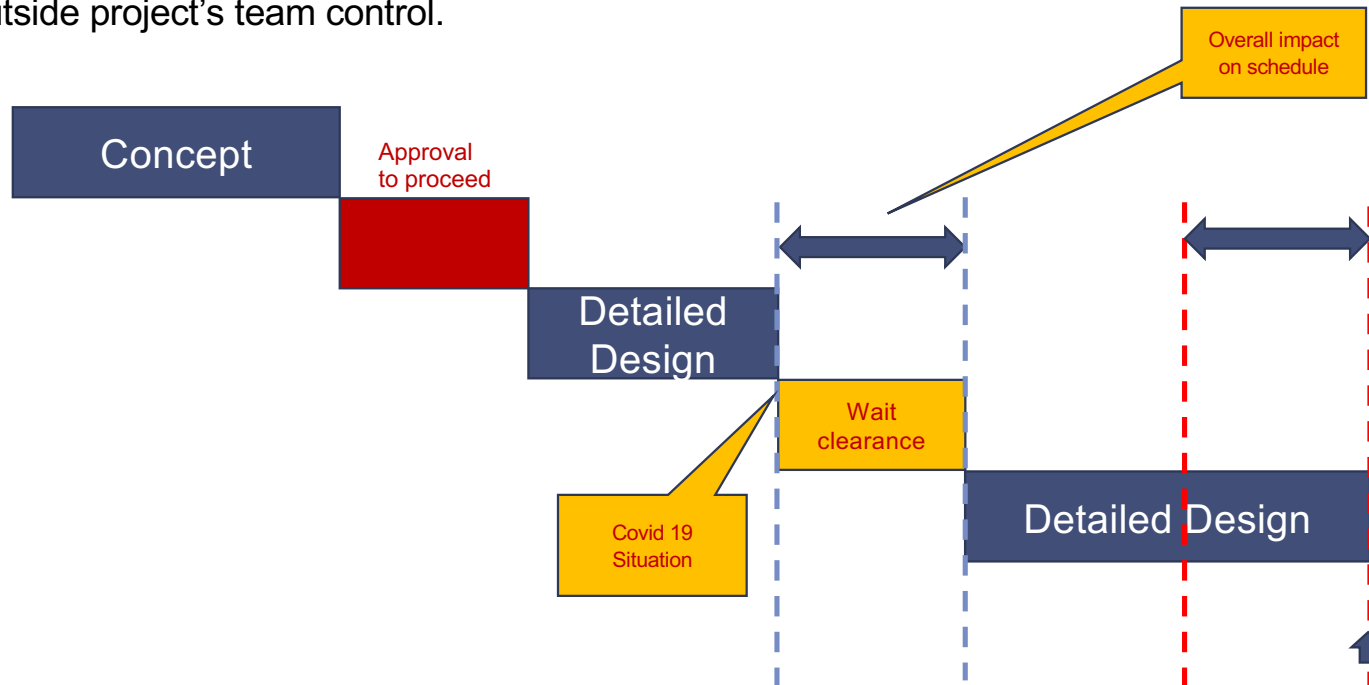


Dependency Determination and Integration

3. External Dependencies:

- Relationship between project activities and non-project activities.
- Outside project's team control.

Example:



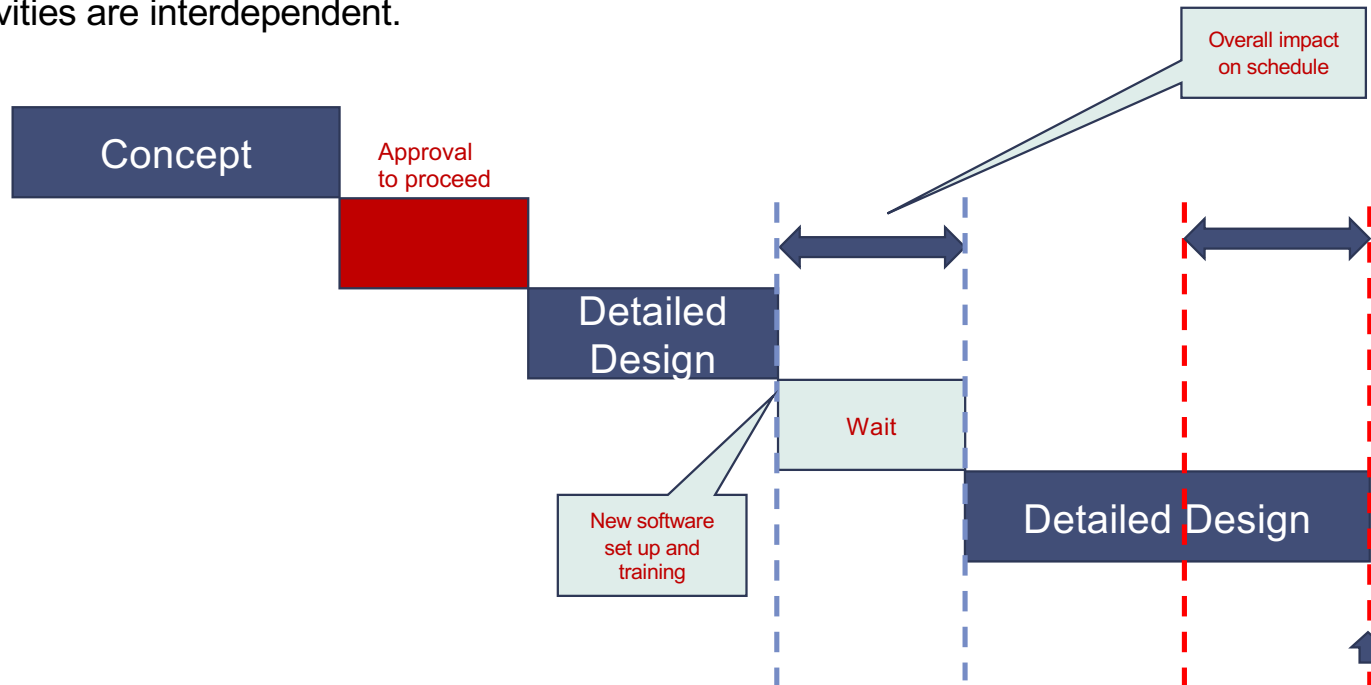


Dependency Determination and Integration

4. Internal Dependencies:

- Inside project's team control.
- Activities are interdependent.

Example:



AI Tool can help



Critical Path Method (CPM) Technique

Project Planning & Scheduling



Critical Path Method (CPM)

This is the **network diagram** method used to **estimate the minimum project duration** and determine the **amount of scheduling flexibility** on the logical network paths within schedule model.



Critical Path Method (CPM)

Activity	A	B	C	D	E	F	G
Immediate Predecessors	--	--	A	A,B	D	C,E	E
Time (weeks)	7	9	12	8	9	6	5

A	ES	EF
t	LS	LF

- A represents Activity Node
- t represents time taken by each activity
- ES Early Start
- LS Latest Start
- EF Early Finish
- LF Latest Finish

$$EF = ES + t$$

$$LS = LF - t$$

$$FLOAT = LS - ES \text{ or } LF - EF$$



Critical Path Method (CPM)

Activity	A	B	C	D	E	F	G
Immediate Predecessors	--	--	A	A,B	D	C,E	E
Time (days)	7	9	12	8	9	6	5

A	ES	EF
t	LS	LF

ES	t	EF
Activity		
LS	Float	LF

PMBok® Guide

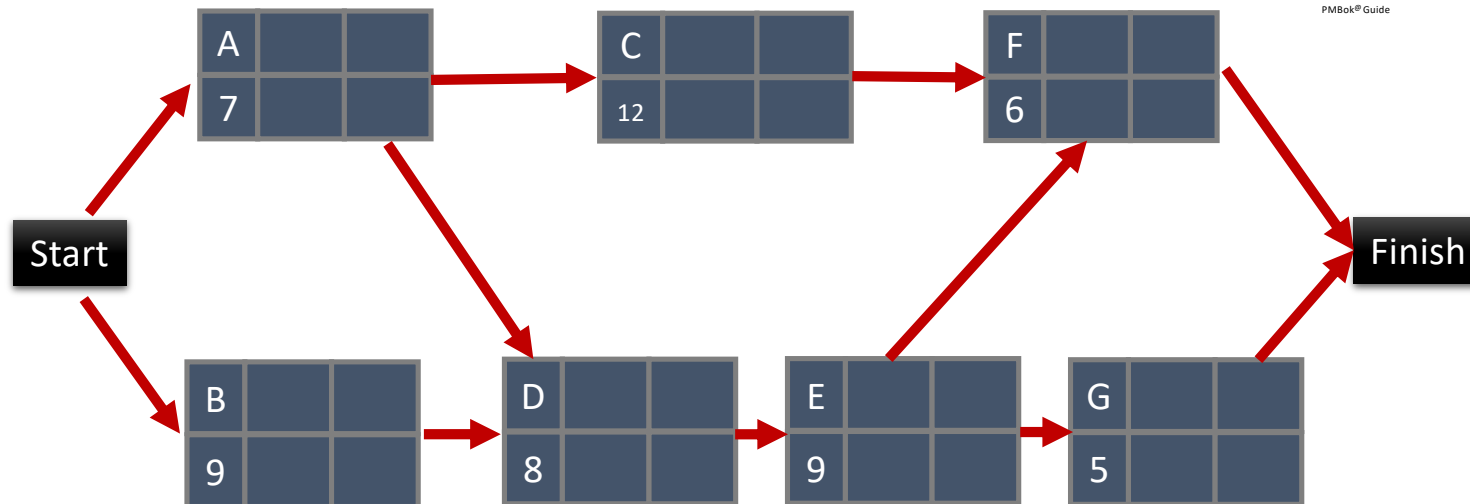
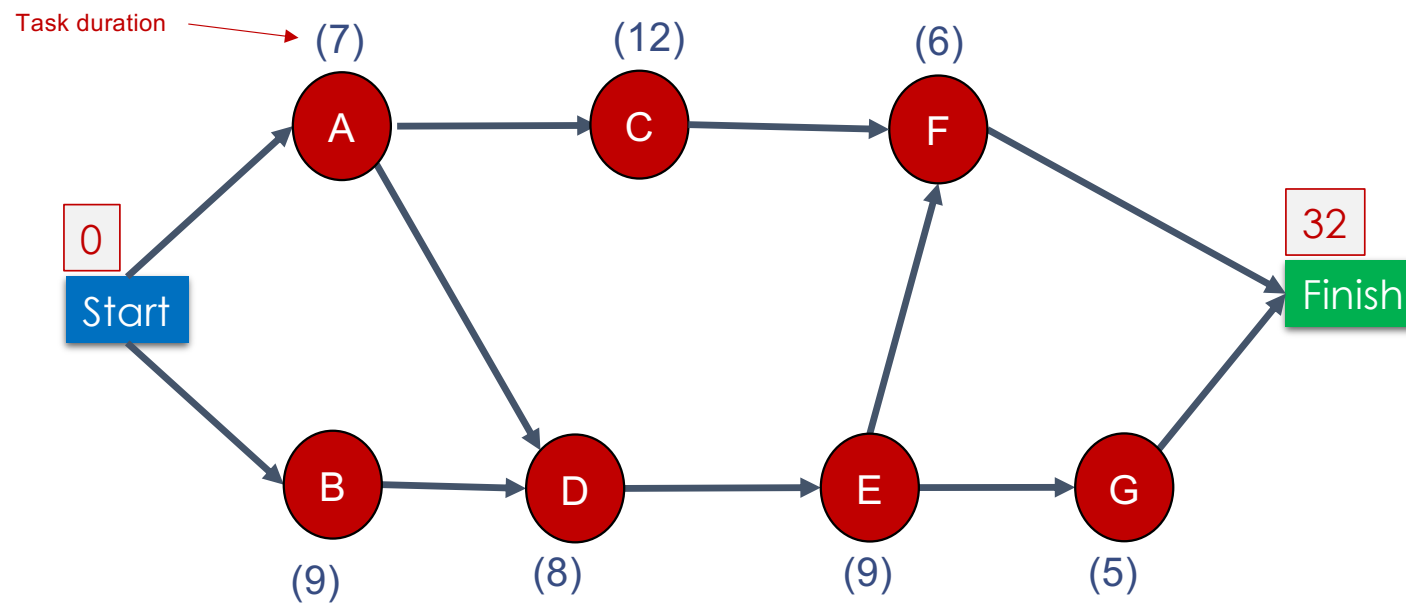




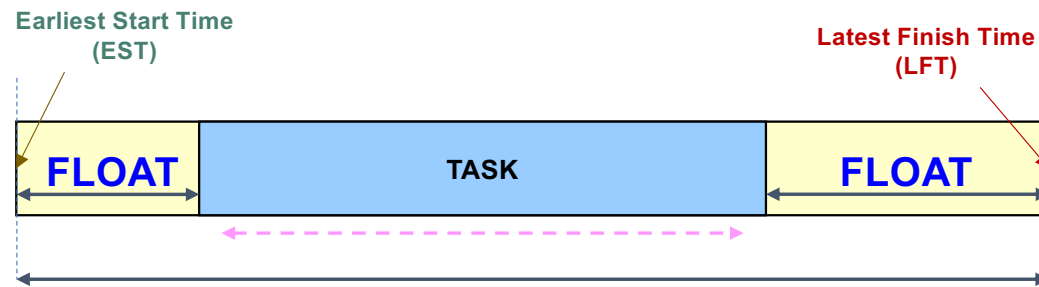
Diagram Representation



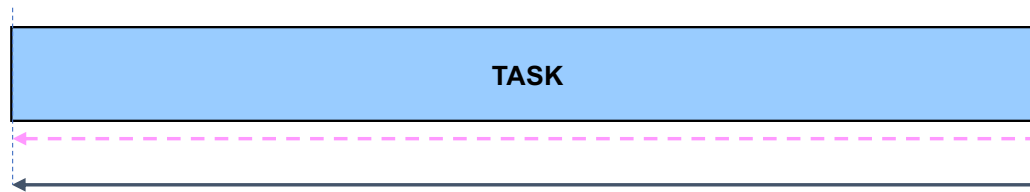


Float

EST, LFT gives an interval which a task can float without affecting the earliest project completion time.



Float is the maximum **permissible time** in a task without affecting project completion.

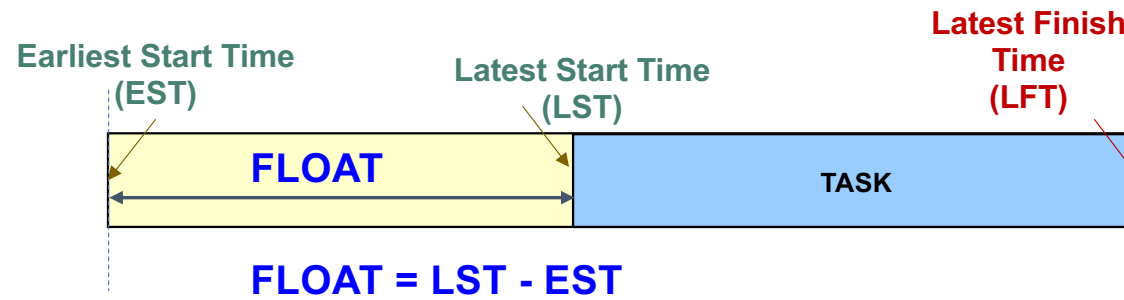
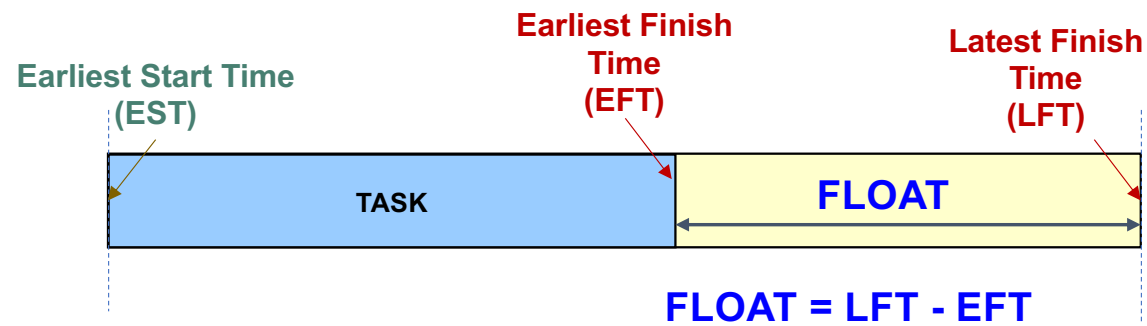


What happens if the task with ZERO Float is delayed?



Float

AI Tool can help

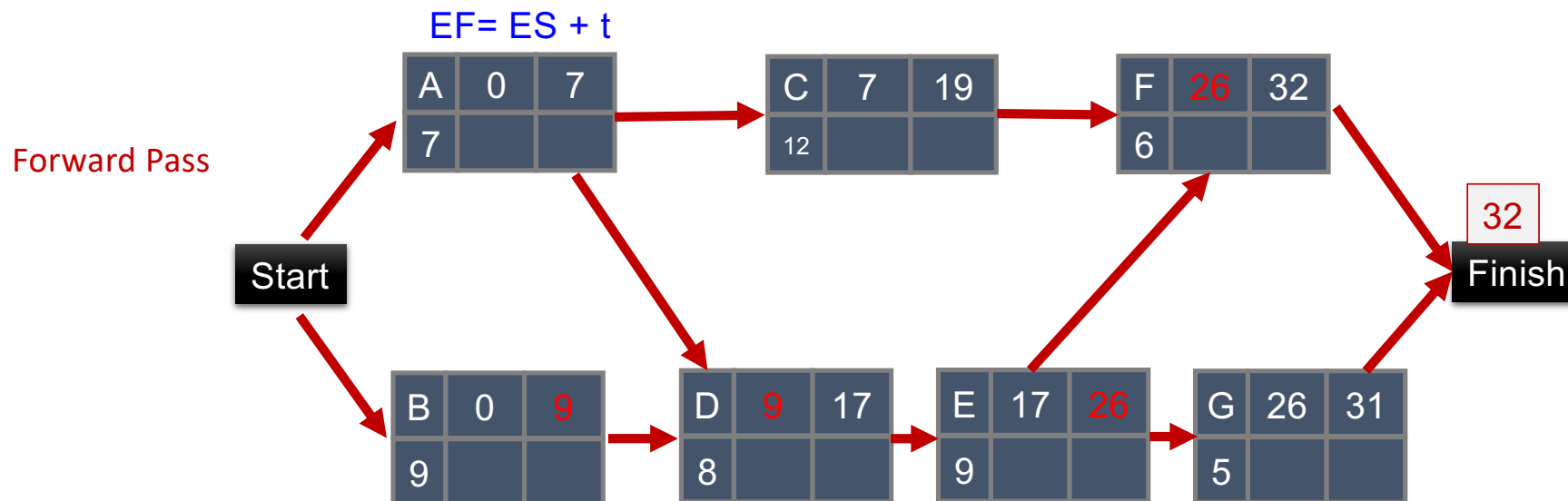




Critical Path Method (CPM)

Activity	A	B	C	D	E	F	G
Immediate Predecessors	--	--	A	A,B	D	C,E	E
Time (days)	7	9	12	8	9	6	5

A	ES	EF
t	LS	LF

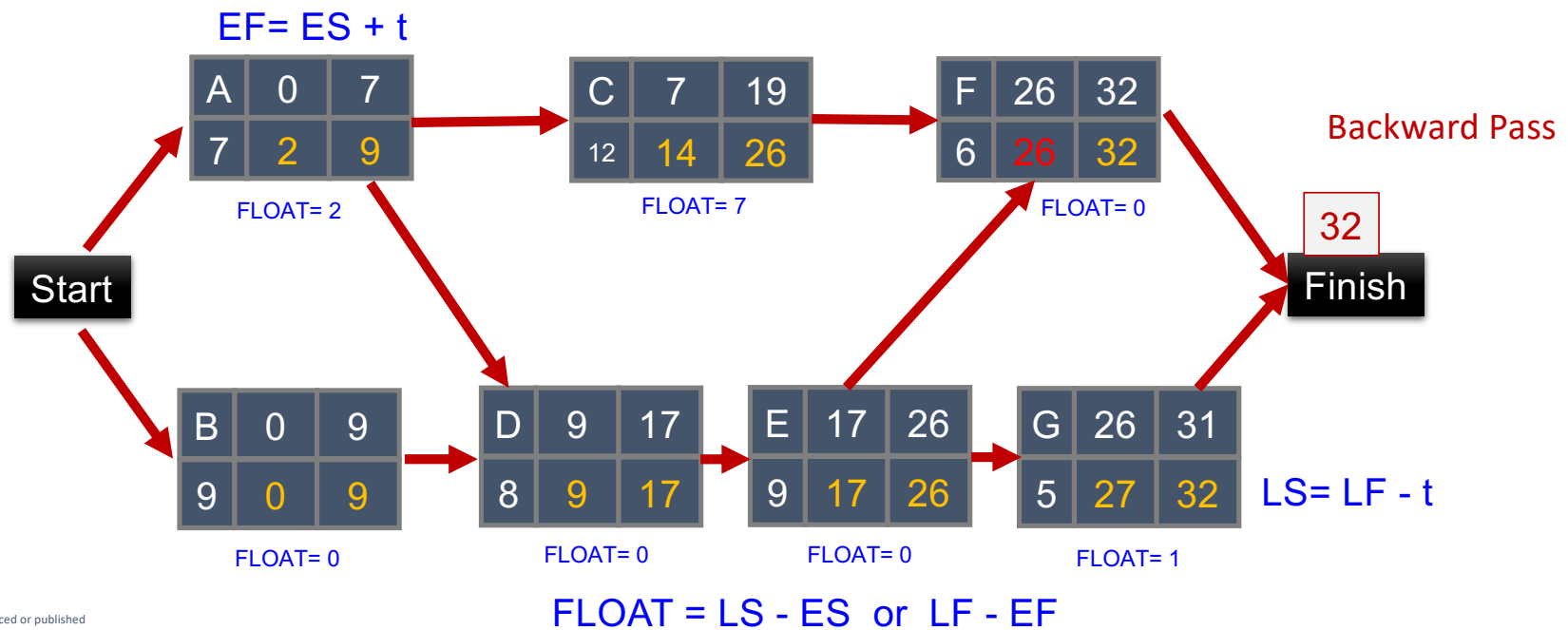




Critical Path Method (CPM)

Activity	A	B	C	D	E	F	G
Immediate Predecessors	--	--	A	A,B	D	C,E	E
Time (days)	7	9	12	8	9	6	5

A	ES	EF
t	LS	LF

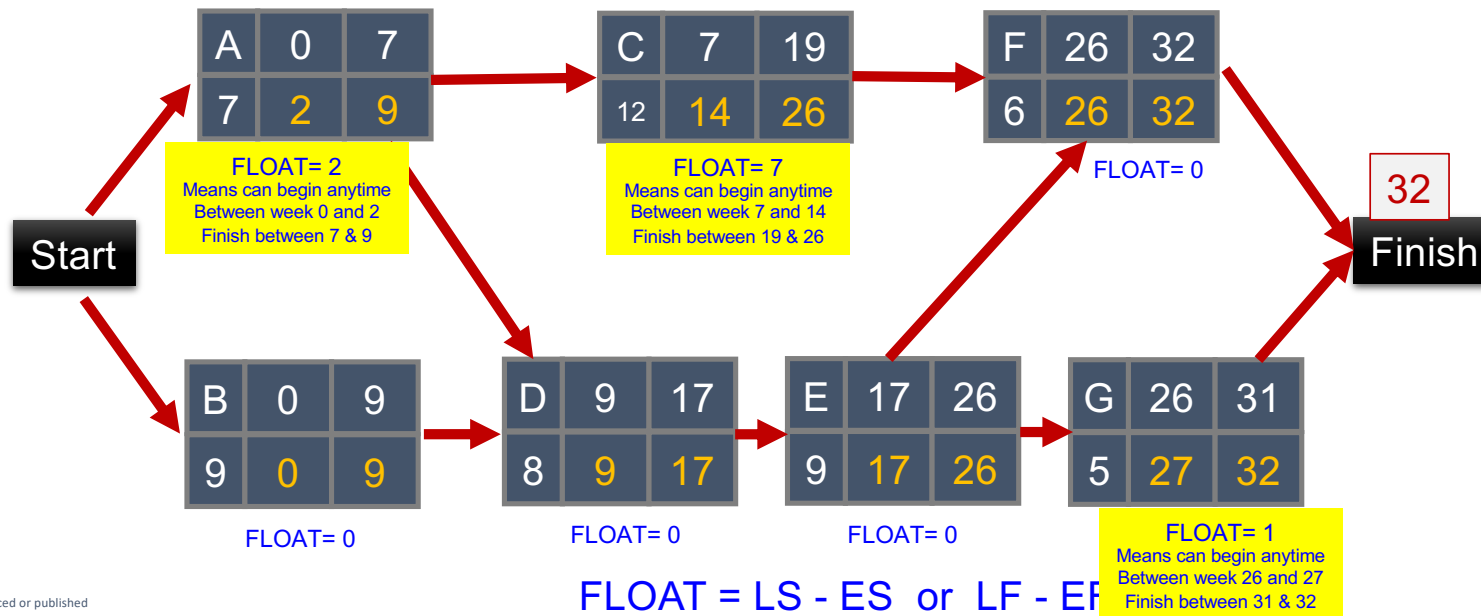




Critical Path Method (CPM)

Activity	A	B	C	D	E	F	G
Immediate Predecessors	--	--	A	A,B	D	C,E	E
Time (days)	7	9	12	8	9	6	5

A	ES	EF
t	LS	LF





Critical Path Method (CPM)

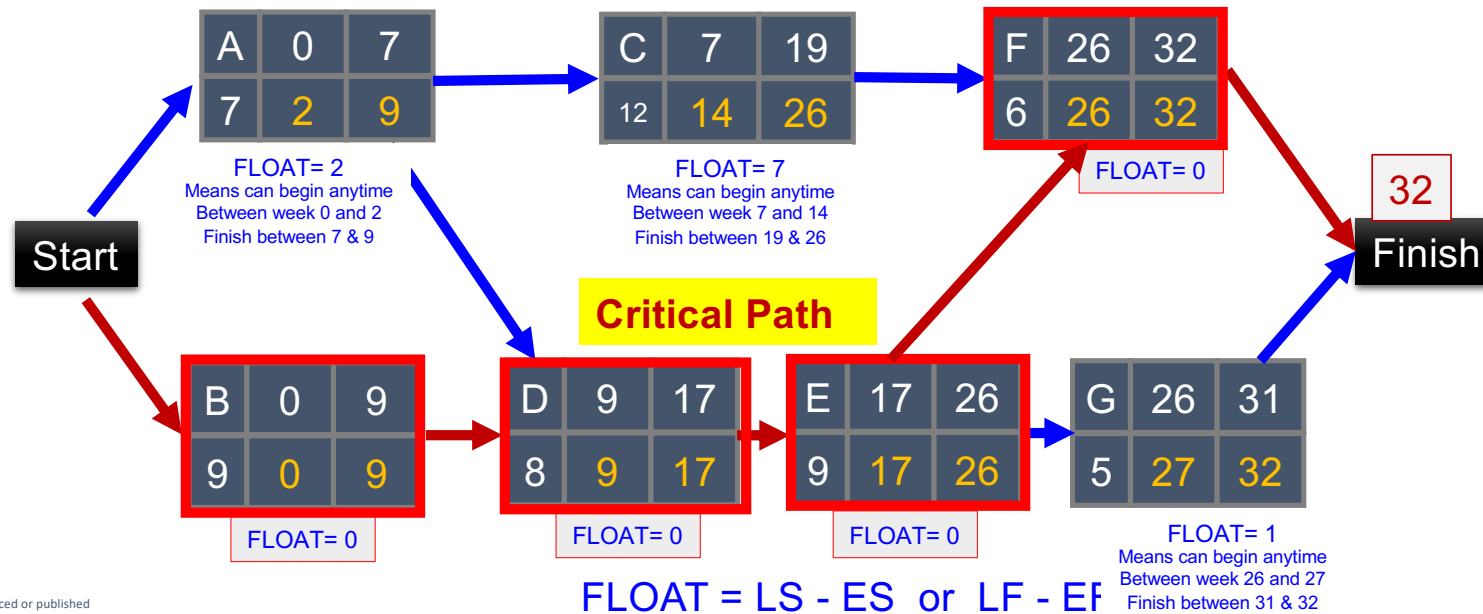
AI Tool can help

Activity	A	B	C	D	E	F	G
Immediate Predecessors	--	--	A	A,B	D	C,E	E
Time (days)	7	9	12	8	9	6	5

A	ES	EF
t	LS	LF

→ Critical Path Link

→ Non-Critical Path Link





Gantt Chart

Project Planning & Scheduling



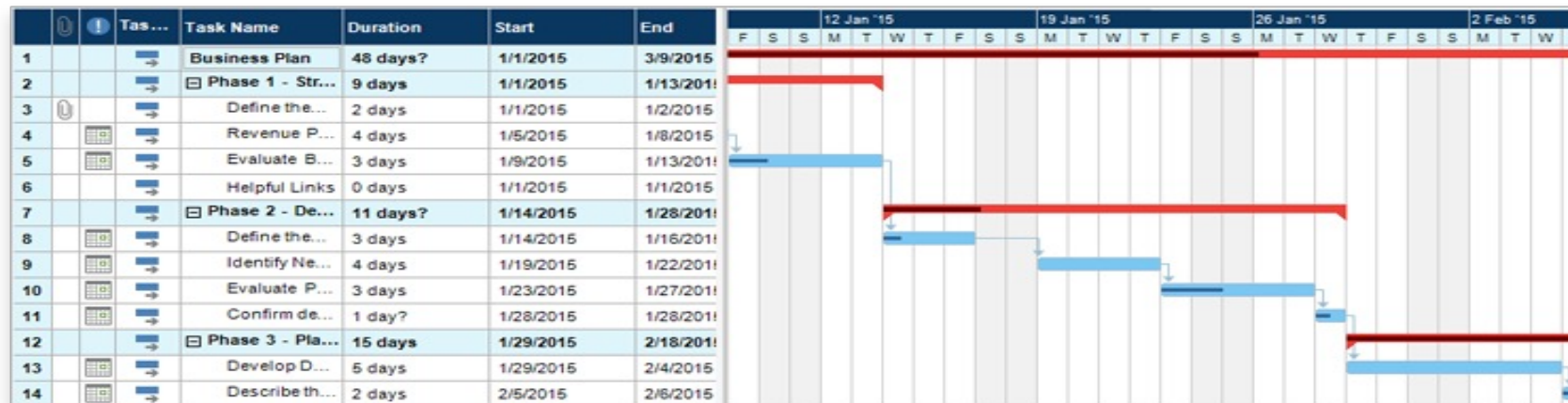
Gantt Chart

A Gantt chart,
commonly used in project management
tool to show activities (tasks or events)
displayed against time.



Gantt Chart

- On the left of the chart is a list of the activities and along the top is a suitable time scale.
- Each activity is represented by a bar; the position and length of the bar reflects the start date, duration and end date of the activity.





Gantt Chart

Gantt chart allows to see:

- Various activities
- Start and finish of each activity
- Duration of each activity
- Activities overlap with other activities
- Critical path activities and Float
- The start and end date of the whole project

AI Tool can help

AI Tool can help

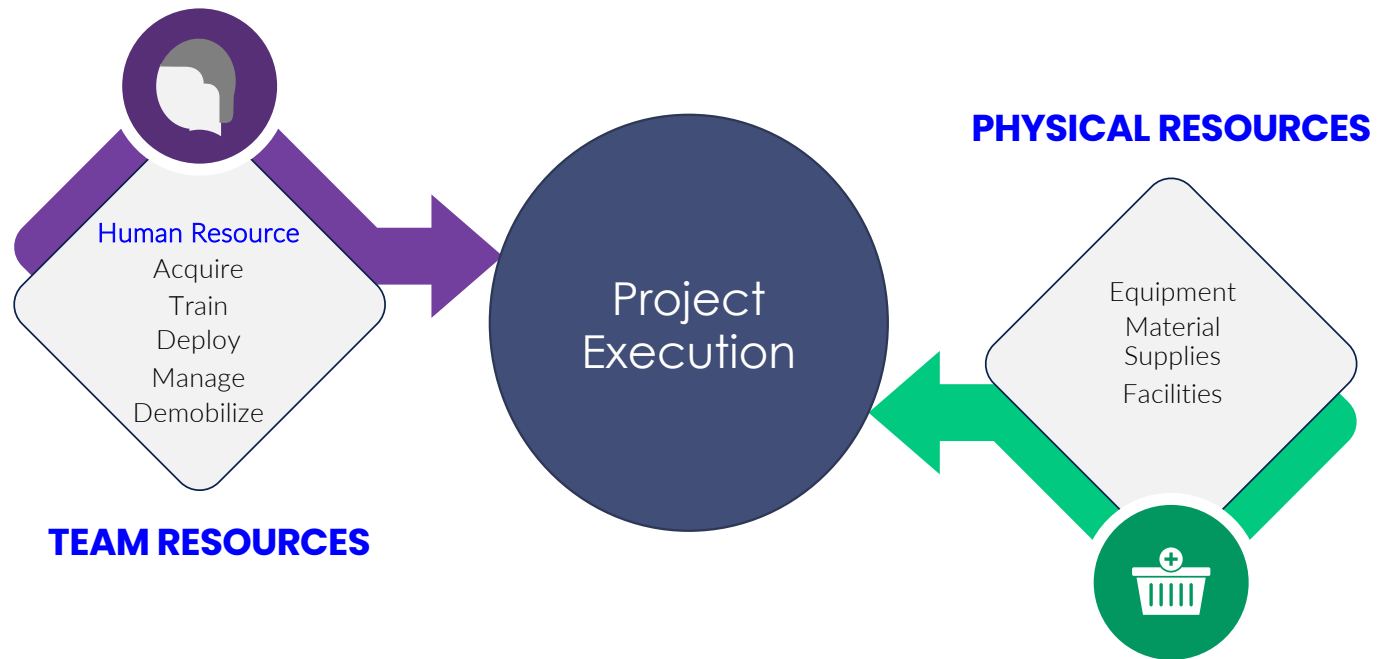


Project Resources

Project Resource Management

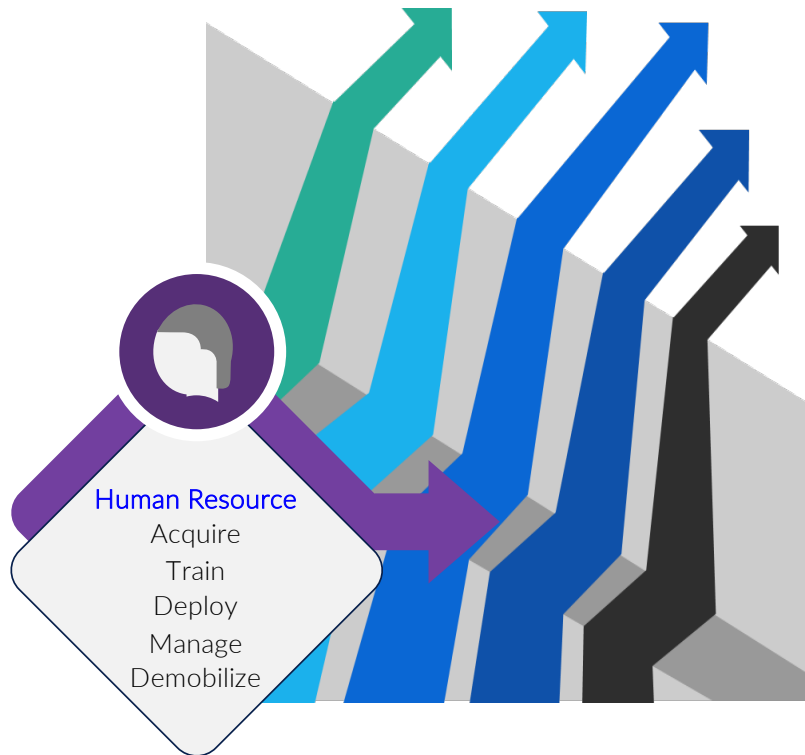


Type of Resources





Project Human Resource Management



TEAM RESOURCES

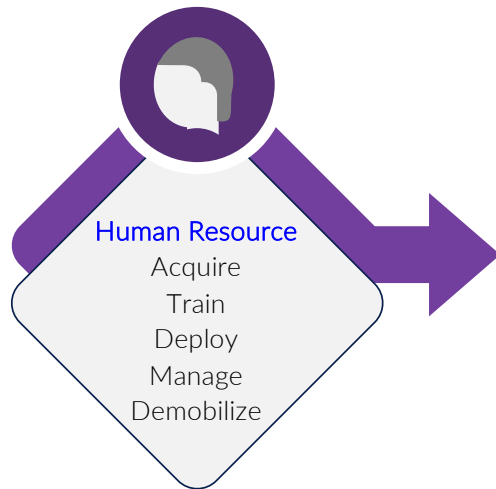
Help to manage Processes

- Plan
- Estimate
- Acquire
- Develop Team
- Manage Team
- Control Resources



Project Human Resource Management

AI Tool can help



TEAM RESOURCES

Various aspects:

- Type of Project
- Team environment
- Geographical locations of team members
- Politics (internal and external)
- Critical issues and any unique features

Question: Can you use “all best resources” on a project?

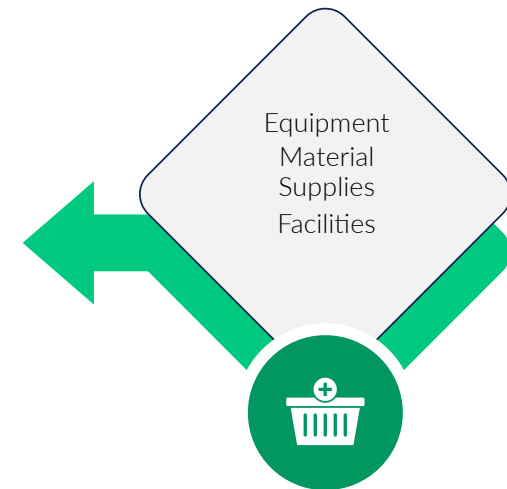


Project **Physical** Resource Management

It is also important and focus on allocation and control of physical resources.

- Material / Software licenses / Cloud
- Equipment
- Supplies
- Facilities
- Transport

PHYSICAL RESOURCES





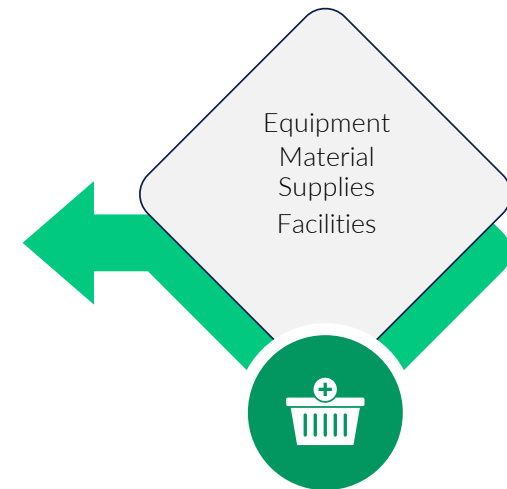
Project **Physical** Resource Management

AI Tool can help

Failing to manage and control physical resources, increase the risk on the project completion.

- **Timely availability** of the equipment is important.
- **Rework** due to non-conformance to the specifications and requirements.
- **Managing quantities.**

PHYSICAL RESOURCES



Question: Do you think project can **order more quantities?**



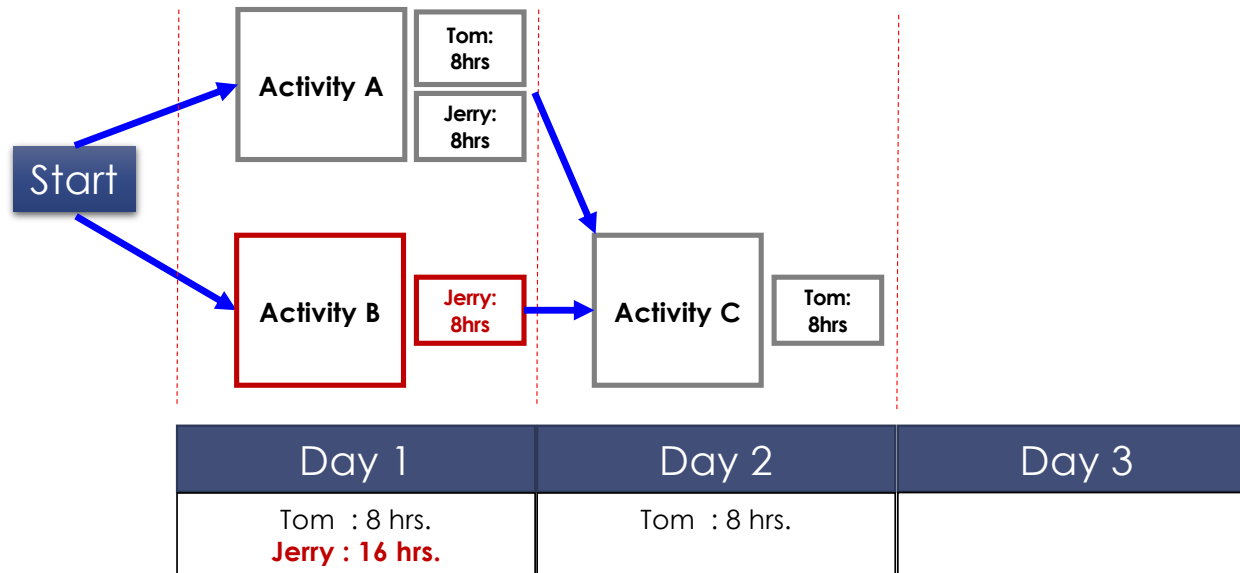
Schedule & Resource Optimization

Project Planning & Scheduling



Resource Optimization Techniques

Activities Before Resource Leveling

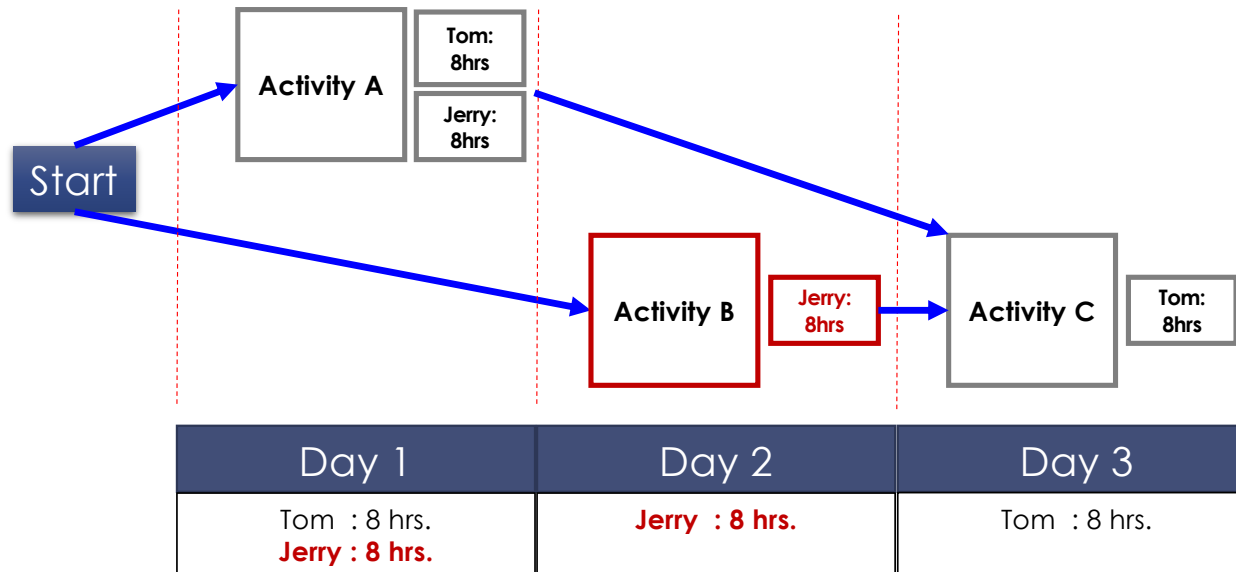




Resource Optimization Techniques

AI Tool can help

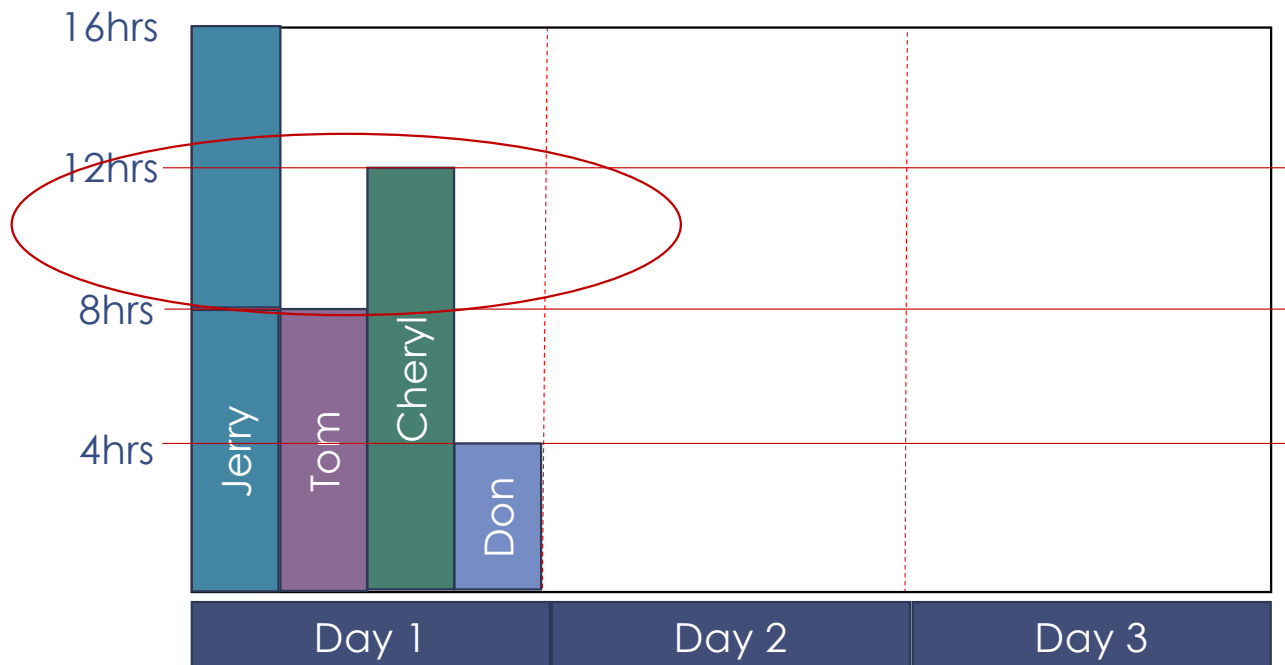
Activities **After** Resource Leveling





Resource Optimization: : Leveling and Smoothing

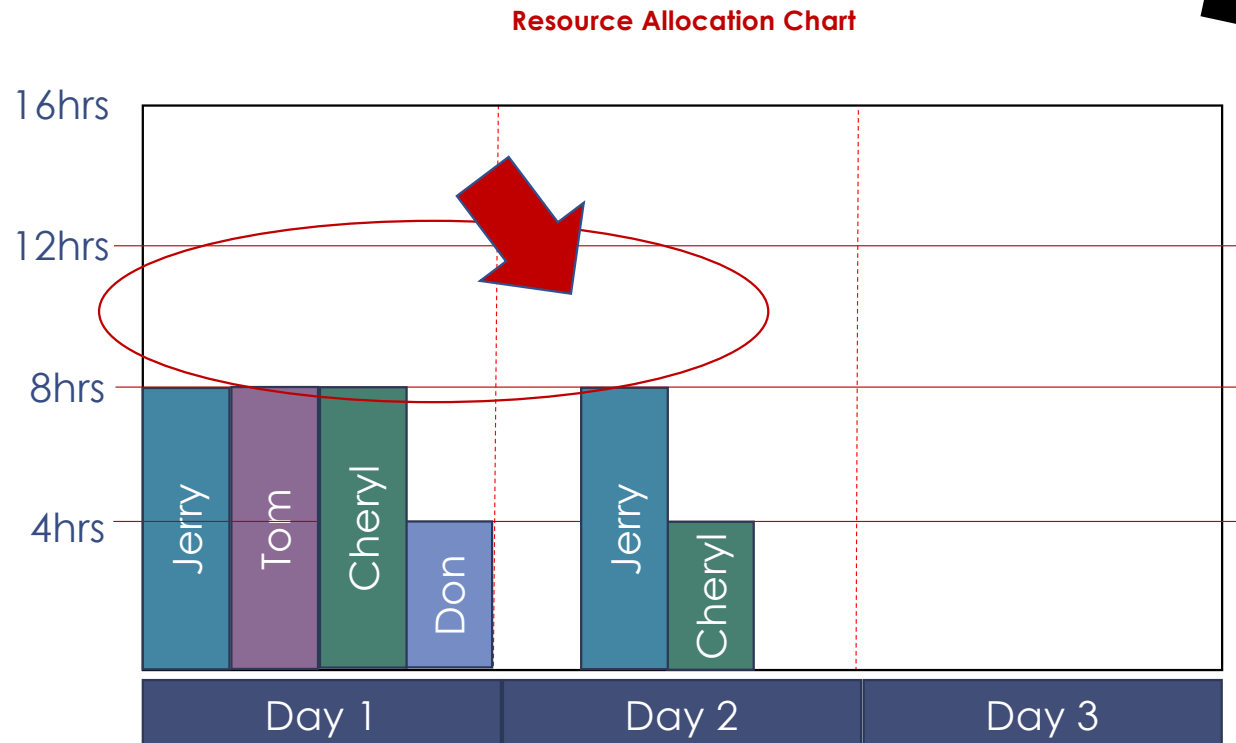
Resource Allocation Chart





Resource Optimization: : Leveling and Smoothing

AI Tool can help





Schedule Optimization: Schedule Compression

AI Tool can help

Techniques used to **shorten the schedule** duration **without reducing scope**, in order to meet schedule constraints, imposed timeline, and other schedule objectives.

1. **Fast Tracking**

Activities and phase delivered in parallel (overlap).

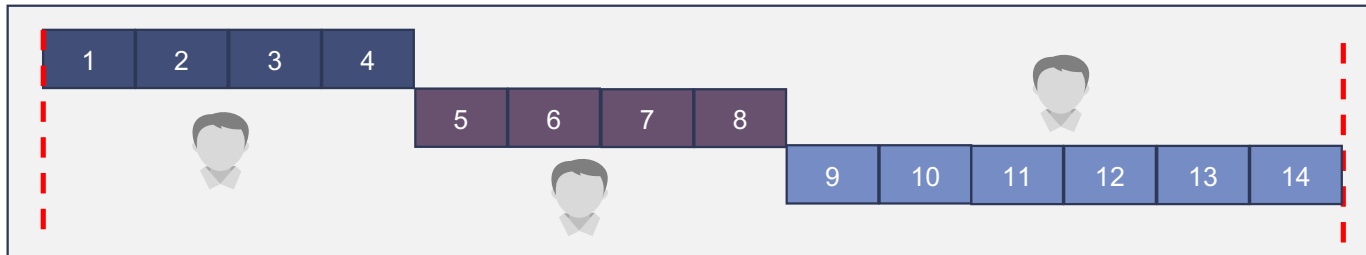
2. **Crashing**

Additional resources to expedite the work. Only to be used for activities under critical path as it may result in cost increase.

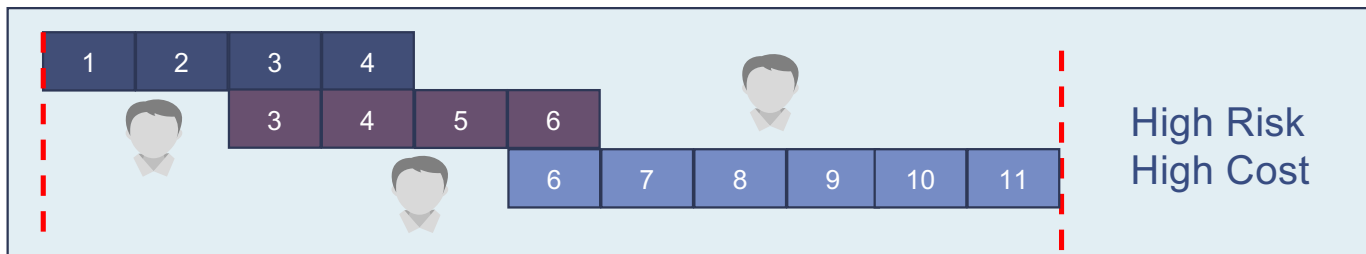


Schedule Optimization: Schedule Compression

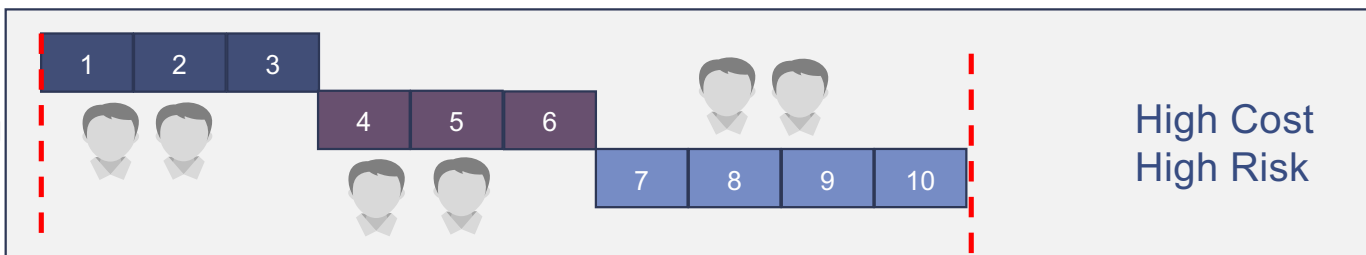
Normal



Fast Tracking



Crashing



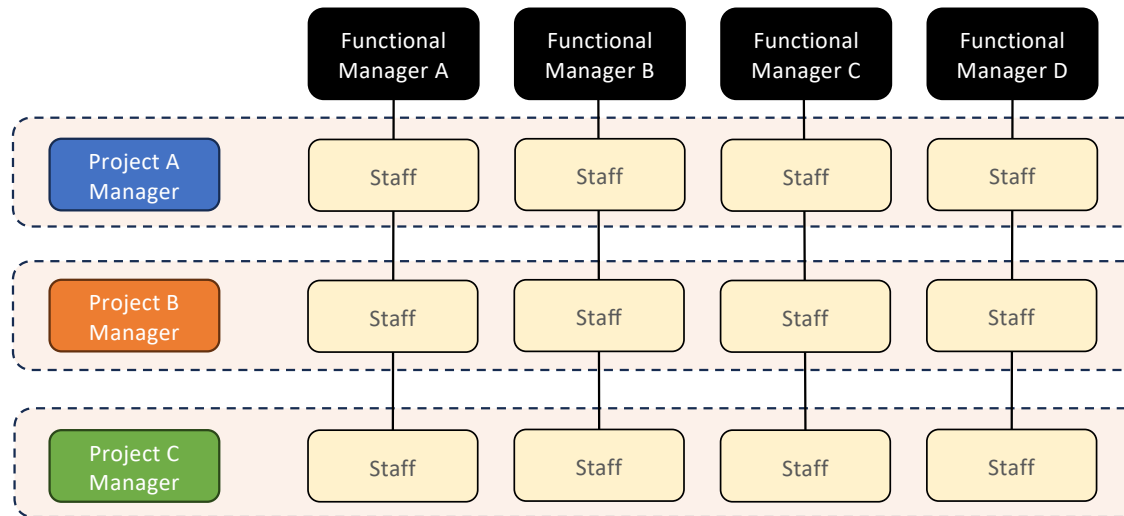


Project Organization

Project Resource Management



Organization Structure (Company Level)

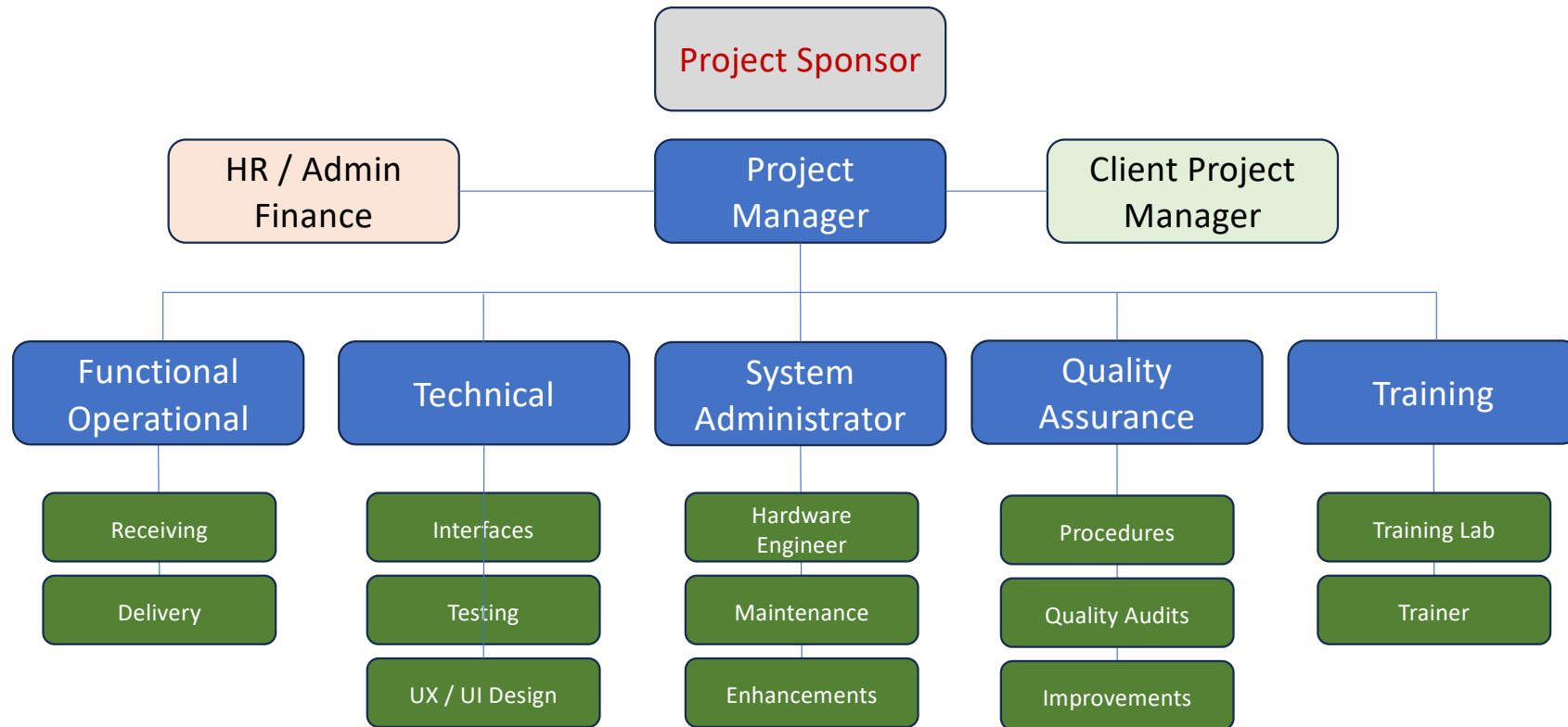


Matrix Organization



Project Organization

Sample





Roles & Responsibilities

Project Resource Management



Project Roles and Responsibilities

Roles and Responsibilities are defined based on:

- Type of Project
- Expertise needed to execute a particular task
- Experience and competencies
- Skills and training
- Willingness

**AI Tool can help
Mapping Skills,
Competencies and
identify training
needs**



Project Roles and Responsibilities

- Role
- Responsibilities
- Authority
- Competency



Project Roles and Responsibilities

AI Tool can help

RACI Matrix is used in large projects.

- Responsible
- Accountable
- Consulted
- Informed

ID	Tasks	Roles	Principal	Project manager	Project member 1	Project member 2	External consultant
1	Formulating project order	C	R				
2	Signing project order	A	A	I	I	I	
3	Situation analysis	I	A	R			
4	Project organization and optimization	I	A	I	I	I	C
5	Work package 1		A			R	C
6	Work package 2		A		C		R

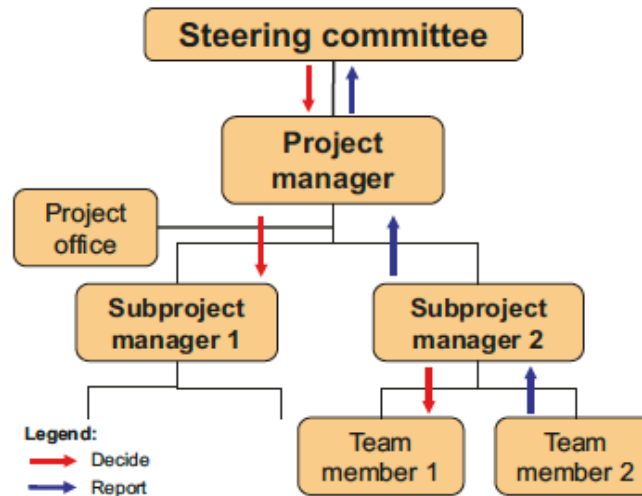
R = Responsible - working on the task
A = Accountable - delegating and approving the work
C = Consulted - typically expert on subject
I = Informed - interested in progress

Source: Project Management for Practice
A Guide and Toolbox for Successful Projects
Daud Alam · Uwe Gühl



Project Roles and Responsibilities

Decision and Reporting paths



Source: Project Management for Practice
A Guide and Toolbox for Successful Projects
Daud Alam · Uwe Gühl



Project Roles and Responsibilities

Points to take note while setting up:

- Manageable size
- Social Mix
- Blend of different competencies
- Experience and skills
- Politics
- Client preferences



Project Staffing

Project Resource Management



Project Staffing

AI Tool can help

Staffing Management Plan

- Staff Recruitment
- Resources Calendar
- Staff Mobilization
- Training Needs
- Recognition and Rewards
- Staff De-Mobilization
- Compliance
- Safety





Project Team Recruitment

AI Tool can help

Project Manager / Team Appointment:

- Similar experience as to the requirement of the project.
 - Type
 - Size
 - Location
 - Familiarity
- Availability
- Willing
- Fits the budget



Project Team Recruitment

How to search the right candidate?

- Internal candidate
 - Local office
 - Other location
- External hire
 - Direct advertisement
 - Referral
 - Agency
 - Consultant

**AI Tool can help to
filter the candidates'
profile**



AI Tools in Project Management

High level Perspective



AI tools in Project Management

AI Tool can help

- Using AI to review the Request For Proposal (RFP).
- Review against previous projects.
- Identify the Risk Profile on this project calling for RFP.
- Review the schedule and purchasing in compliance to the policies.



Machine Learning in Project Management

AI Tool can help

Estimating

- Analysis of past projects and create a new estimate for future projects.
- Challenge is to build the accurate data set



Machine Learning in Project Management

AI Tool can help

Scheduling

Can build up the schedule using the data on:

- Resource availability
- Task Dependencies
- Task Durations
- Project Constraints

- Parallel working
- Fast tracking
- Additional resources

Number of alternative scenarios possible for the given information & priorities



AI tools in Project Management

AI Tool can help

- Schedule optimization.
- Schedule changes to the project within our control or outside and responding to the changes.
- Shop fabricated became overbooked and need some intervention.
- Review the scope and optimize etc.
- If need to be de-scope etc.
- Shipping impact
- Linear regression model solution (descope and re-scheduling can be done).



AI tools in Project Management

AI Tool can help

- Linear progression approach can help.
- Baseline schedule is available in different formats or software.
- Any changes will have some impact on the schedule.
- Creating a workflow to review the impact is a mini-project.
- Change request model can be embedded with approval processes.



Prediction and forecasting

Project Planning & Scheduling



Prediction and forecasting

- Earned Value Management covers prediction and forecasting.
- Any changes may have an impact on schedule, cost, risk, and resources.



Case Studies

AI in Project Management (Not in AI Project)

AI in Project Management



Case Study A

**AI in Project
Management**

Case Study: Development of Business Case

Scenario: GenAI is used to generate the business case documentation.

Tools used: LLM via ChatGPT

Concerns:

- Confidential information data could not be used.
- Need human intervention to review the outputs.
- Output was not in the existing format available in the company.
- Project Manager still needed time to review the outcome.

Solutions:

- Inhouse LLM (ChatGPT similar) model will help to resolve confidentiality issues.
- Implement human-in-the-loop review for AI alerts
- Provide training to build trust among workers



Case Study B

**AI in Project
Management**

Case Study: Generation schedule with critical path identified.

Scenario: GenAI is used to generate schedule with highlighted critical path.

Tools used: LLM via ChatGPT

Concerns:

- Confidential information data could not be used.
- Need human intervention to review the outputs.
- Project Manager still needed time to review the outcome.

Solutions:

- Implement human-in-the-loop review for AI alerts
- Provide training to build trust among workers



Case Study C

*AI in Project
Management*

Case Study: Automatic summarizing the notes of meeting in google drive.

Scenario: AI Agent workflow is used to generate the output.

Tools used: Make, Google Drive, Jotform, LLM via ChatGPT

Concerns:

- Paid version of ChatGPT developer version was used to generate APIs.
- Jotform subscription.

Solutions:

- Good workflow which could be implemented easily.



Case Study D

**AI in Project
Management**

Case Study: Automatic summarizing the invoices from different suppliers.

Scenario: AI tools are used to generate expected outcome.

Tools used: Expensify, Altyrex, Excel

Concerns:

- Expensify and Altyrex are paid account to utilize to max.
- Learning curve.

Solutions:

- Good workflow which could be implemented easily.



Case Studies

AI Projects (Not in Project Management)

AI Projects



Case Study A1

AI Project

Case Study: Predictive Maintenance in Manufacturing

Scenario: AI is used to predict equipment failure before it happens to avoid unplanned downtime.

Ethical concerns:

- Reliability: False positives lead to waste; false negatives lead to danger
- Transparency: Operators may not trust a black-box system
- Accountability: Who is responsible when a machine fails after an AI alert?

Creative Solutions:

- Build interpretable models with visual diagnostics
- Implement human-in-the-loop review for AI alerts
- Provide training to build trust among workers



Case Study B1

AI Project

Case Study: Structural Safety AI

Scenario: Civil engineers use AI to detect stress and predict failure in bridges and buildings.

Ethical concerns:

- Explainability: Regulatory bodies require justification for decisions
- Dataset limitations: Lack of failure examples can limit accuracy
- Safety: Engineers must never defer responsibility to the AI

Creative Solutions:

- Combine AI with physics-based modeling
- Use explainable AI methods for transparency
- Maintain final decision-making with licensed professionals



Case Study C1

AI Project

Case Study: Energy Optimization in Chemical Engineering

Scenario: ML optimizes energy usage, reducing waste and emissions in a chemical plant.

Ethical concerns:

- Pushing systems to the limit could sacrifice safety
- Collection of sensitive operational data
- Environmental justice: Are negative externalities properly considered?

Creative Solutions:

- Impose safety bounds in optimization routines
- Use encrypted, anonymized data for analysis
- Regular audits for environmental compliance and fairness



Case Study D1

AI Project

Case Study: Autonomous Vehicles

Scenario: An AI system controls a self-driving car to reduce accidents.

Ethical concerns:

- Crash dilemmas: How should it decide between two harmful outcomes?
- Bias in perception systems (e.g., skin tone, age)
- Legal accountability after accidents

Creative Solutions:

- Program safety envelopes and default-safe behavior
- Diversify training data to minimize bias
- Use black-box logs to ensure transparency in decision-making



Case Study E1

AI Project

Case Study: Smart Cities and Privacy

Scenario: AI is used in traffic control, surveillance, and city planning.

Ethical concerns:

- Surveillance: Citizens unaware of being tracked
- Data retention and usage without consent
- Algorithmic bias in policing or resource allocation

Creative Solutions:

- Engage communities in system design and governance
- Apply privacy-preserving data practices
- Implement fairness audits for public-facing AI decisions



Student Exercise 2

- Perform a search or collate your own personal experience to identify at least 1 case study where you think AI is used in Project Management.
 - Any industry.
 - Any location.
 - Any type of project (IT, Construction, Pharma, etc.)

Prepare your response and submit via
Submission Link Provided Separately.



End of class

